Roots

About the model

What is Roots?

Roots applies a method called **Disease Course Mapping**, originally developed to track how Parkinson's disease progresses over time, to autism severity scores instead. Given a handful of visits for a child, it estimates where that child sits on a shared developmental timeline and how quickly they are moving along it, then predicts their likely trajectory on each item of the **CARS2-ST** severity scale.

How it was built

The underlying statistics come from Poulet and Durrleman (2023, *Statistics in Medicine*), who built this method for a Parkinson's scale with 59 items (MDS UPDRS). Roots reuses the same model for the CARS2-ST scale, which has 15 items.

- No real longitudinal CARS2-ST dataset was available, so the model was trained on **realistic synthetic data** (300 simulated children), generated to match published score ranges and the three factor structure of CARS2-ST, as published by Campbell et al. (2026): Social Communication, Restrictive/Repetitive Behavior and Sensory, and Emotional/Behavioral Dysregulation.
- For each item, the model learns a **population average curve**: how scores typically change with age.
- For each child, it then estimates two personal dials from their actual visit scores, **pace** (faster or slower than average) and **timing** (earlier or later start), and uses those to shift the population curve into a personalized one.

Pros and cons

Pros

- Reuses a rigorous method validated on real neurodegenerative disease data, applied here to a new domain.
- Naturally handles irregular visit schedules, few visits per child, and ranked (ordinal) scores.
- Gives an interpretable personal summary (pace and timing), not just a single severity number.
- Captures the CARS2-ST's published three factor severity structure.

Cons and limitations

- Trained on synthetic, not real patient, data: a proof of concept, not a validated clinical tool.
- Assumes severity moves in **one direction** per item, so it cannot show a child worsening on one item while improving on another within the same run.
- At smaller sample sizes the model struggles to cleanly separate the three severity factors, a known limitation found during testing.
- Needs real longitudinal CARS2-ST data to properly validate before any clinical use.

How to use and interpret it

- Enter the child's age and CARS2-ST item scores for each visit in the **Trajectory Explorer**.
- After at least one visit, the tool shows **Developmental pace** (1.00x means exactly average pace; higher or lower means faster or slower than the population average) and **Timing offset** (how many years earlier or later this child started, relative to average).
- Pick any item from the dropdown to see this child's personalized curve (solid teal line) against the population average (dashed grey line), with recorded visits shown as dots.
- Treat the curves as an illustration of how the method works, not as a prediction to act on. This is a demonstration built on synthetic data, not a diagnostic or clinical decision support tool.

Reference: Poulet, P. E. and Durrleman, S. (2023). Multivariate disease progression modelling with longitudinal ordinal data. *Statistics in Medicine*.
| CARS2-ST factor structure: Campbell et al. (2026). | CARS2-ST item content, copyright Schopler et al., 2010 (WPS), is not reproduced in this tool.